

Foundations of Natural Language Processing

Peking University, 2026

Assignment 4

For Graduating Students: Due Monday, June 29, 2026 by 23:59
For Other Students: Due Monday, July 6, 2026 by 23:59

1 Directions

Please read the instructions carefully to ensure that you receive full credit for this assignment.

Late submission policy. This policy does not apply to graduating students.

- Late homework will have 5% deducted from the final grade for each day late.
- No submission will be accepted after July 13 2026, one week after the due date for non-graduating students.
- Your homework submission time will be based on the time of your last submission to Course.
- Please be aware that all late submissions, regardless of the reason, will be penalized according to the late submission policy. It is your responsibility to plan accordingly and submit your homework on time.

2 Task Description

Recent frontier language models have demonstrated substantial progress on mathematical and scientific reasoning tasks. However, such performance often depends on very large model scale, extensive pretraining, and high inference cost. Smaller general-purpose models remain more prone to failures in specialized knowledge recall, quantitative calculation, unit conversion, symbolic manipulation, and multi-step physical or chemical reasoning.

This assignment focuses on how a fixed small language model can be systematically augmented to approach the problem-solving ability of stronger models. The base model for all leaderboard submissions is **Qwen3-8B**. Your task is to develop methods that use tools, retrieval, verification, structured prompting, or other reproducible system-design techniques to improve it on university-level physics and chemistry fill-in-the-blank questions. In other words, the central research question is how external scaffolding can help a smaller model close part of the capability gap.

2.1 Data

The released evaluation dataset contains 230 university-level questions: 112 physics questions and 118 chemistry questions. A small number of entries include answers as examples.

Each data entry may contain the following fields:

Field	Description
<code>id</code>	Problem identifier.
<code>field</code>	Discipline, either <code>physics</code> or <code>chemistry</code> .
<code>question</code>	The problem statement.
<code>answer</code>	The answer field. Most released test entries contain <code>null</code> .
<code>subfield</code>	The subfield of the problem.
<code>theorem</code>	A possible physics law or principle, when applicable.
<code>unit</code>	The unit of the answer, when applicable.

The evaluator is designed with math-verify¹, and supports LaTeX-style mathematical answer comparison; therefore you do not need to worry about harmless differences in equivalent mathematical notation.

Allowed external knowledge sources. You may use **only** the following textbooks as external knowledge bases:

- *University Physics*, Volumes 1–3, by Ling et al. (2016).
- *Atkins’ Physical Chemistry*, by Atkins, de Paula, and Keeler (2023).

The `.tex` versions of these textbooks are uploaded to <https://course.pku.edu.cn/>. We also provide `.pdf` versions on <https://disk.pku.edu.cn/link/AA026EBB450B3C4F3C8F75E62BAB23AA53>

2.2 Required Baseline

You must implement and report the following baseline:

- **Model:** Qwen3-8B.
- **Prompting method:** direct chain-of-thought prompting.
- **Reference accuracy:** approximately 50%.

If your baseline result differs substantially from the reference range, please debug your implementation, prompt formatting, answer extraction, and submission format before developing more advanced methods.

¹<https://github.com/huggingface/Math-Verify>

2.3 Possible Methods

You may use any prompting or system-design strategy, as long as your method follows the data, model, and resource restrictions in this instruction. The methods should be designed to augment Qwen3-8B rather than replace it with a stronger model. Examples include, but are not limited to:

- self-consistency;
- retrieval-augmented generation;
- agentic pipelines;
- symbolic, numerical, or coding tools.

You must clearly describe the complete pipeline in your report. The analysis should be rigorous and evidence-based: it should explain why your method works or fails, quantify improvements over the direct Qwen3-8B baseline, identify major error types, compare the behavior on physics and chemistry, and discuss representative success or failure cases involving retrieval, tools, or verification.

Apart from that, you also have to write your answer extraction script. Make sure to extract your answers as Python strings, as our evaluator only accepts this type. It should be noted that all submitted answer strings **must not** contain descriptions, equations, or units. Here are some examples of valid and invalid answer expressions:

Valid	Invalid
3.14	The answer is 3.14°
$\frac{22}{\sqrt{7}}$	$x = \frac{1}{2}mv^2 = 8.32$
3e8	3e8 m/s
6.67×10^{-11}	The answer is the gravitational constant

2.4 Kaggle Leaderboard

Join the Kaggle leaderboard for this assignment using the link announced by the teaching team.

- Kaggle link: <https://www.kaggle.com/t/b1c9bad212594238b92178430c655b67>
- Please change your Kaggle team name to your student ID. Otherwise, we cannot identify your leaderboard score and you may receive zero credit for the leaderboard component.
- Please actively use the leaderboard system to evaluate your methods. You can submit prediction results up to 10 times every 24 hours. Do not delay your experiments until the last few days.
- Follow the Kaggle competition page for the required submission file schema. Your submission should contain one final answer for each problem id.

2.5 LLM API Requirement

In this project, you will likely conduct experiments with LLM APIs. To ensure a fair comparison between students, please use the Qwen3-8B API from <https://bailian.console.aliyun.com/> for experiments submitted to the Kaggle leaderboard. Note that search models are not limited if your pipeline contains searching modules.

Your report may discuss comparisons with other models, but the results submitted to Kaggle **must** be produced using Qwen3-8B.

3 Submission

Submit all required materials to <https://course.pku.edu.cn/> before the deadline. Your submission should include:

1. **Report.** The report should describe your method, experimental setup, implementation details, leaderboard results, baseline comparison, analysis, etc.
2. **Code.** Submit code that can reproduce your reported Kaggle submission. Include scripts for model calls, retrieval or tool use, **answer extraction**, submission-file generation, and **original reasoning traces**.
3. **Prediction file.** Submit the final prediction file (.csv) that corresponds to your selected Kaggle submission. The columns should include "id" and "answer". A sample submission file is uploaded to the Kaggle competition and Course website for reference.
4. **Running instructions.** Include a short README or command list explaining how to run your code and reproduce the submitted predictions.

Reproducibility and verification. To prevent cheating, the teaching team will verify submitted code using the Qwen3-8B model. If the results reproduced from your code differ substantially from your Kaggle submission, your final grade may be affected.

4 Grading Criterion

We will grade your project based on the following criteria:

1. **Report (80%).** In the report, you should:
 - describe your method clearly;
 - report the implementation details of your method;
 - evaluate your method on the leaderboard;
 - compare your method with the required baseline to demonstrate the effectiveness of your approach;
 - analyze the experimental results in depth.

2. **Implementation (5%).** We will check whether you have submitted all required materials. The submitted code should be reproducible.
3. **Leaderboard (15%).** Part of the grade will be determined by your ranking on the leaderboard.

5 Contact

If you have general questions about Assignment 4, please contact:

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